

Renal reabsorption of glucose is facilitated by sodium-glucose linked cotransporters (SGLTs), which are secondary active transport proteins located in the apical membrane of the proximal tubular cells in the kidney. Glucose transported out of the proximal tubule via SGLTs is returned to circulation through the peritubular capillaries. SGLT activity determines the transport maximum of glucose T_m , the maximum rate at which the kidney can reabsorb glucose.

Excretion of glucose in the urine (glycosuria) occurs when the glomerular filtration rate (GFR) of glucose exceeds the T_m . Glycosuria is a symptom of diabetes mellitus, a disease marked by hyperglycemia (elevated serum glucose). Uncontrolled hyperglycemia can damage the nerves that innervate urinary tract musculature, resulting in urinary retention (inability to empty the bladder completely) or incontinence (involuntary urine leakage).

Canagliflozin is an SGLT inhibitor used in the treatment of diabetes. The effect of canagliflozin on renal glucose reabsorption was tested in a glucose infusion experiment in patients with type 2 diabetes mellitus (T2DM). First, researchers measured the participants' fasting serum glucose levels (average ~80 mg/dL) and then began a continuous intravenous infusion of a glucose solution. The rate of glucose infusion was adjusted such that plasma glucose concentration steadily rose to 300 mg/dL. Plasma and urine glucose concentrations were measured at 30-minute intervals. The experiment was later repeated in the same patients following administration of canagliflozin (100 mg once a day for 7 days). The results of both trials are shown in Figure 1.

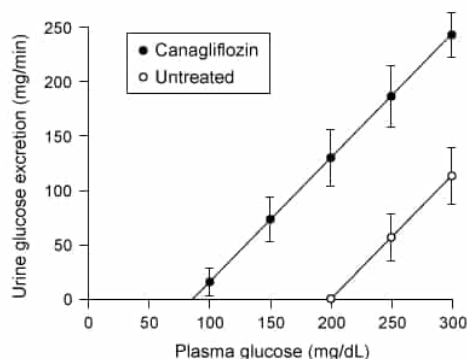


Figure 1 Mean urine glucose excretion vs. blood plasma concentration in patients with T2DM, with and without canagliflozin treatment

Researchers then estimated the new T_m of glucose following canagliflozin treatment (Figure 2).

Canagliflozin was found to also lower fasting plasma glucose in individuals with T2DM. In addition to the blood pressure reduction that typically accompanies canagliflozin therapy, common adverse effects included weight loss, increased urinary frequency, and excessive thirst.

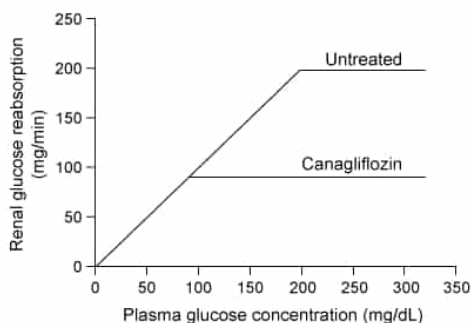


Figure 2 Mean rate of renal glucose reabsorption vs. plasma glucose concentration in patients with T2DM, with and without canagliflozin treatment

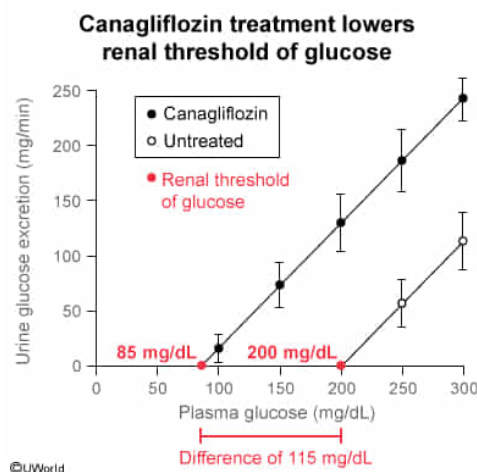
The critical concentration of plasma glucose necessary to induce glycosuria is termed the renal threshold of glucose. The renal threshold of glucose in T2DM patients treated with canagliflozin compared to the threshold in untreated T2DM patients is:

- ☐ A. higher by approximately 75 mg/dL. [5%]
- ☐ B. higher by approximately 175 mg/dL. [5%]
- ✓ ☒ C. lower by approximately 115 mg/dL. [88%]
- ☐ D. lower by approximately 300 mg/dL.

Correct

88% answered correctly

Explanation:



Urine formation represents the net result of the processes of filtration and reabsorption in the [kidney](#). According to the question, the *renal threshold of glucose* is the plasma glucose concentration at which glycosuria (ie, excretion of glucose in the urine) occurs. Therefore, at this threshold, the GFR of glucose must *exceed* the rate at which this filtered glucose can be reabsorbed by the kidney, which would lead to excretion of glucose in urine (ie, urine glucose excretion would surpass 0 mg/min).

Figure 1 shows the average amount of glucose excreted in the urine (y-axis) as a function of plasma glucose levels (x-axis) for both untreated and canagliflozin-treated patients with T2DM. According to Figure 1, urine glucose excretion exceeded 0 mg/min in the untreated condition at a plasma glucose concentration of about 200 mg/dL. Urine glucose excretion exceeded 0 mg/min in the canagliflozin condition at about 85 mg/dL. Therefore, **T2DM patients** treated with **canagliflozin**, when compared with untreated T2DM patients, had a **renal threshold of glucose** that is **lower** by approximately **115 mg/dL**.

(Choices A and B) An increase in the renal threshold of glucose would produce a graph in which the canagliflozin-treated line is shifted *to the right* of the untreated line. In Figure 1, the canagliflozin-treated line is shifted *to the left* of the untreated line, indicating a *decrease* in the renal threshold of glucose.

(Choice D) According to Figure 1, the renal threshold of glucose in patients with T2DM was approximately 200 mg/dL. Canagliflozin treatment cannot possibly decrease the renal threshold of glucose by 300 mg/dL because this would result in a negative plasma glucose concentration.

Educational objective:

A threshold is a set value that must be surpassed for a particular event to take place.